

Taiwan Cancer Registry

Sex-Specific Cancer Atlas — 78,441 Patients · 37 Sites · 2003–2020

Overview

Sex is the strongest binary covariate in the Taiwan Cancer Registry. The VAE identified two orthogonal sex-linked axes:

- z4 – UADT/field-cancerization (rank-biserial=−0.621, male)
- z0/z5 – Hormonal/gynecologic (rbi=+0.163/+0.409, female)

These axes map to distinct carcinogenic exposures:

- Male → betel nut + tobacco (UADT) + HBV/metabolic syndrome (GI)
- Female → endogenous hormones (breast, endometrial, ovarian)

Key Findings (all FDR<0.05)

Site sex-OR spectrum:

- Male-dominant: 23 sites
- Female-dominant: 1 sites
- Sex-neutral: 13 sites

Strongest male site: C12 pyriform OR=40×

Strongest female site: C50 breast OR=0.002×

Age × sex interaction:

- UADT males present 5–9yr YOUNGER than UADT females
(C06: Δ=−9yr, C02: Δ=−7yr, C15: Δ=−5yr)
- C50 breast: males 15yr OLDER than females (incidental/late-onset)

Multi-cancer rate:

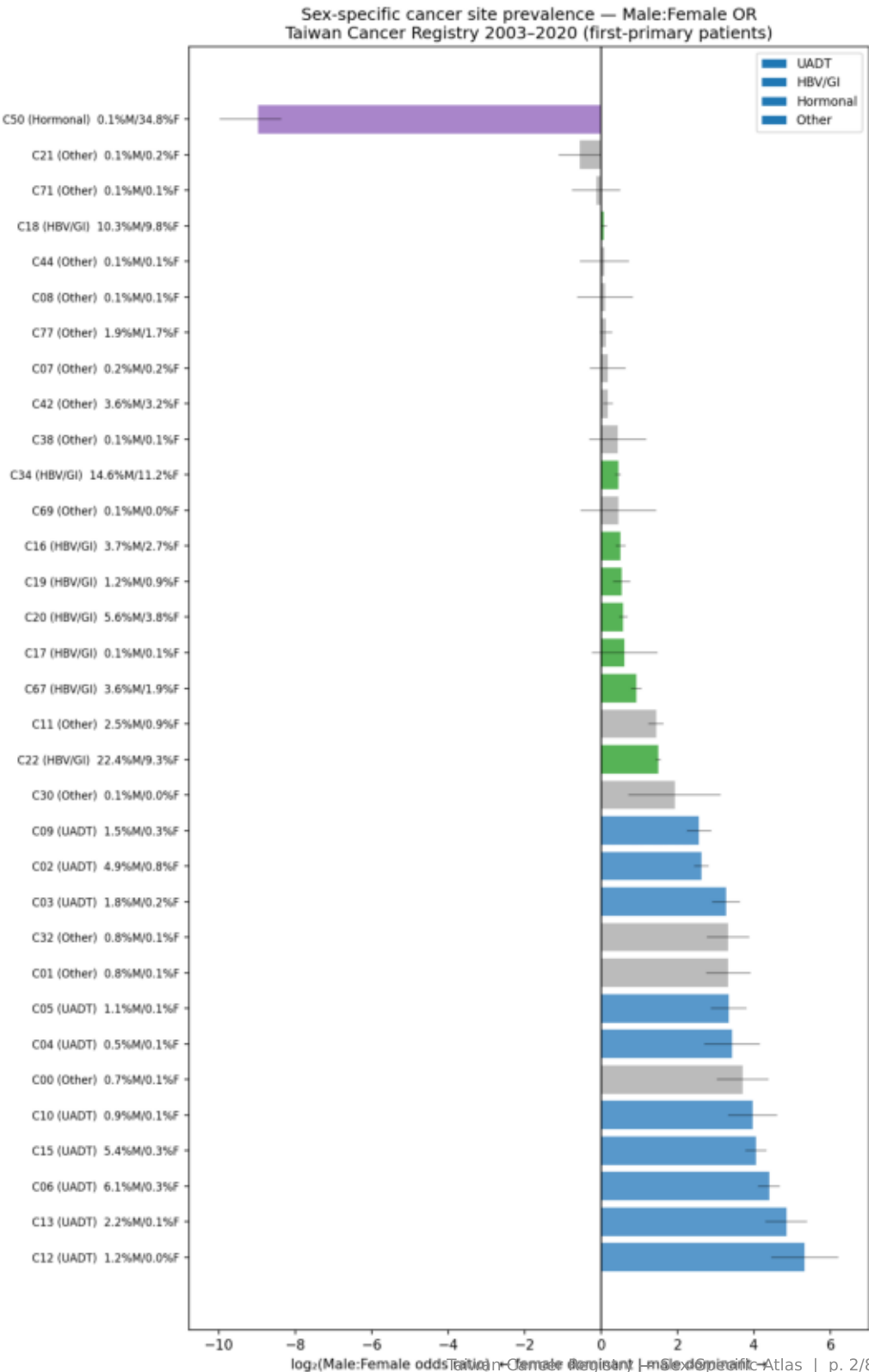
- Males: 6.8% vs Females: 3.4% (2.0× difference)
- UADT males: 14.9% vs UADT females: 7.7% (axis-specific)

Survival:

- Log-rank M vs F: p<0.001; males worse overall

Sex-specific cancer prevalence — Male:Female OR forest

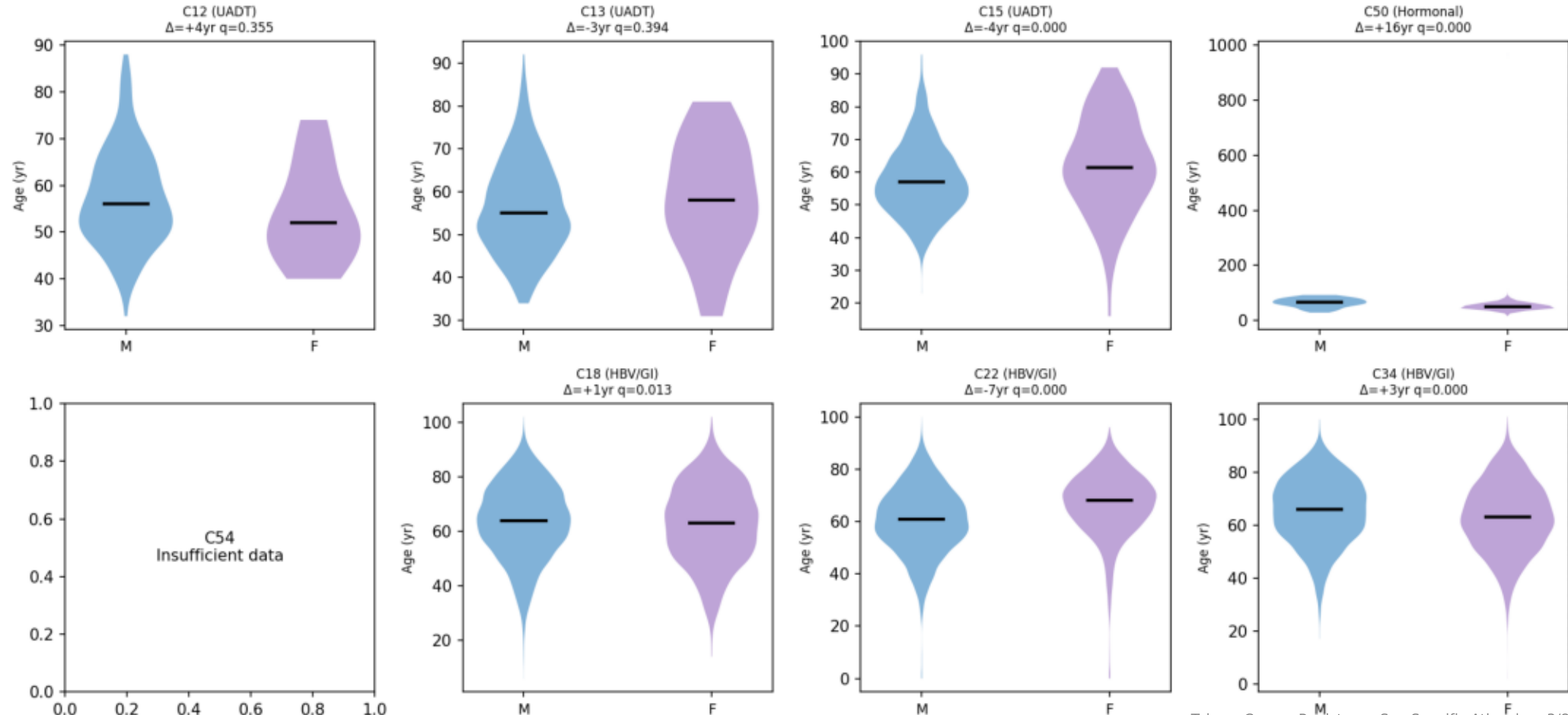
Male:Female OR by site — log₂ scale (all 37 sites)



Age at first diagnosis by sex — UADT males present younger

Age at first diagnosis by sex — key sites (blue=male, purple=female)

Age at first diagnosis by sex — key cancer sites
(blue=male, purple=female; Δ = median male – female)



Age × sex differences by site (FDR<0.05, top 12)

Site	Axis	M median	F median	Δ (M–F)	q FDR
C69	Other	64	46	+18	0.0344
C50	Hormonal	66	51	+16	0.0000
C00	Other	56	69	-13	0.0037
C03	UADT	56	66	-10	0.0001
C08	Other	61	51	+10	0.0117
C06	UADT	52	61	-9	0.0000
C02	UADT	51	58	-7	0.0000
C22	HBV/GI	61	68	-7	0.0000
C15	UADT	57	62	-4	0.0004
C34	HBV/GI	66	63	+3	0.0000
C16	HBV/GI	66	63	+3	0.0004
C67	HBV/GI	67	70	-2	0.0096

Age × sex — biological interpretation

UADT sites: males present YOUNGER

C06 oral cavity: −9yr C02 tongue: −7yr
C15 esophagus: −5yr C03 floor of mouth: −10yr

Interpretation: heavy betel nut + tobacco exposure begins in adolescence/early adulthood in Taiwan males. Earlier cumulative exposure → earlier cancer onset. Female UADT patients likely have lower betel/tobacco exposure → present later.

C22 liver HCC: males −7yr younger

Consistent with higher HBV carrier rate in males (male HBV clearance is lower → higher chronicity).

C50 breast: males +15yr older

Male breast cancer is rare, late-onset, typically post-60yr in Taiwanese men. Female breast cancer peaks 40–55yr (younger than Western populations).

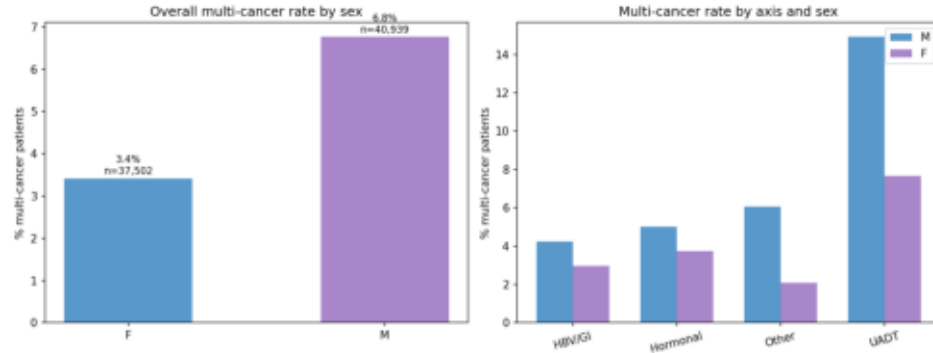
Clinical implication:
Screening programmes should be sex-stratified and use different age-at-initiation thresholds.

Males 2x more likely to develop sequential cancers

Multi-cancer rate by sex and axis

Multi-cancer rate by sex (overall + by axis)

Multi-cancer rate by sex — males 2x more likely to develop sequential cancers



Multi-cancer rate by sex — key finding

Overall:

Males: 6.8% Females: 3.4% Ratio: 2.0x

By carcinogenic axis:

UADT M: 14.9% vs F: 7.7% Ratio: 1.9x

HBV/GI M: 4.2% vs F: 3.0% Ratio: 1.4x

Other M: 6.0% vs F: 2.1% Ratio: 2.9x

Interpretation:

The 2x male excess in multi-cancer rate is not simply explained by UADT field cancerization (the largest contributor) — Other-axis males also show 2.9x excess, suggesting a systemic sex difference in multi-primary risk.

Possible mechanisms:

1. Betel nut + alcohol + tobacco act synergistically across multiple anatomical sites (not only UADT)
2. Immune surveillance differences between sexes: higher female cytotoxic T-cell activity may suppress nascent second primaries more effectively
3. Hormonal influence on DNA repair capacity

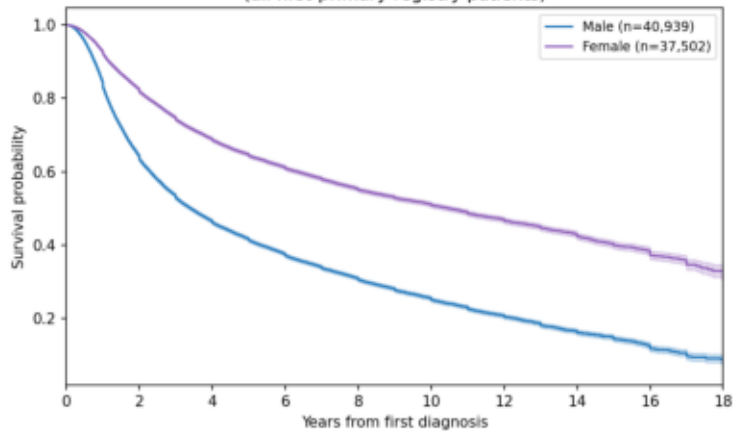
Note: hormonal-axis males n=40 only (prostate + lung) — UADT dominates male carcinogenesis.

Survival by sex — overall and UADT-specific

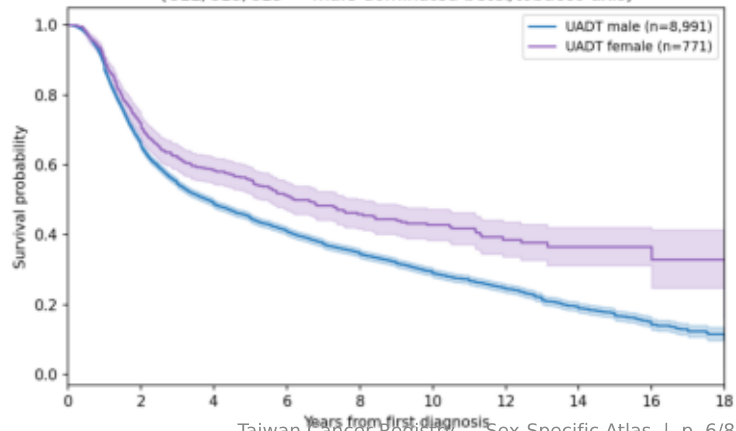
Kaplan-Meier survival by sex (overall + UADT subset)

Kaplan-Meier survival by sex

Overall survival by sex
(all first-primary registry patients)



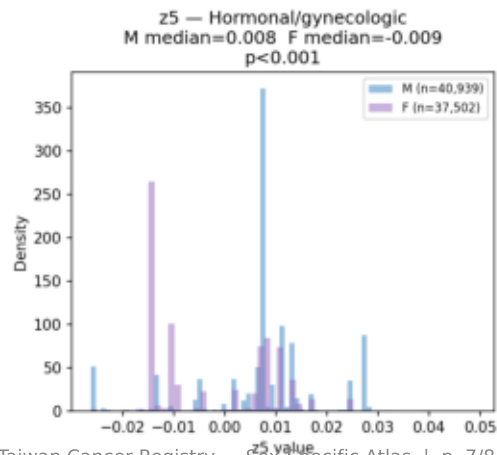
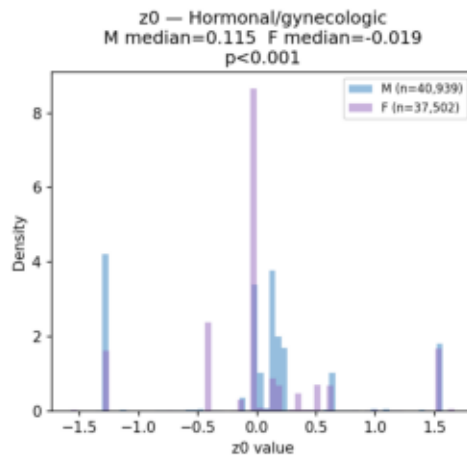
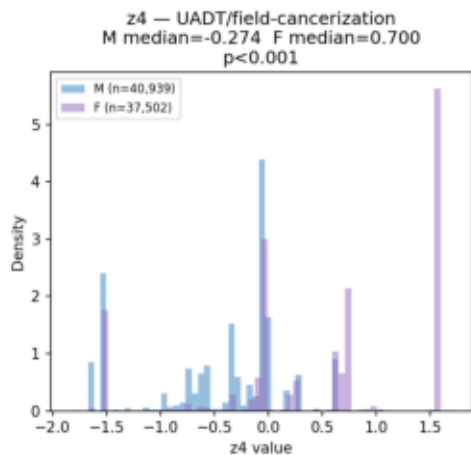
UADT-site patients: survival by sex
(C12/C13/C15 — male-dominated betel/tobacco axis)



VAE latent axes encode sex — z4 male-shifted, z0/z5 female-shifted

VAE active axis values by sex — latent space sex separation

VAE active axis values by sex — latent space sex separation
(z4=UADT: male-shifted; z0/z5=Hormonal: female-shifted)



Conclusions and Limitations

Sex-Specific Cancer Atlas — Taiwan Cancer Registry 2003–2020

Principal Conclusions

1. The Taiwan cancer registry is strongly sex-stratified.
Of 37 sites, 23 are male-dominant, 1 are female-dominant, and 13 are sex-neutral – all $FDR < 0.05$.
2. Two orthogonal sex-linked carcinogenic axes:
MALE → betel/tobacco UADT (z_4 , $rbi = -0.621$) + HBV/GI
FEMALE → hormonal/gynecologic (z_0/z_5 , $rbi = +0.409$)
These axes explain the bulk of sex heterogeneity in the registry.
3. UADT males present 5–9yr younger than females for the same site.
Consistent with earlier-onset betel nut exposure in male adolescents.
Screening programmes should use sex-specific age-at-initiation.
4. Males have 2× the multi-cancer rate of females (6.8% vs 3.4%).
Excess is not confined to UADT – 'Other' sites show 2.9× excess.
Possible mechanisms: carcinogen synergy, immune sex differences.
5. Survival is significantly worse in males (log-rank $p < 0.001$).
Partly explained by UADT predominance (high-mortality sites) and later-stage presentation.

Limitations

No betel nut/tobacco/alcohol exposure data – sex differences are attributed to exposure axes by ecological inference only.
No hormonal data (parity, OCP, HRT) for female hormonal axis.
Registry records first-primary site per patient; rare non-primary sex assignment errors possible for sex-exclusive sites.
Multi-cancer rate may be partially confounded by follow-up duration (males have higher baseline mortality → shorter FU window for detecting second cancers).

Next Steps

Sex-stratified survival models for each axis (Cox with sex × axis interaction)
Age-standardised incidence rates by site and sex (vs general population)
Link sex-specific multi-cancer risk to Transformer surveillance calendar
(male UADT patients warrant more aggressive 6-monthly endoscopy)